

(MIS)ATTRIBUTING THE CAUSES OF AMERICAN JOB LOSS THE CONSEQUENCES OF GETTING IT WRONG

DIANA C. MUTZ*

Abstract What difference does it make if people attribute the loss of manufacturing jobs to trade as opposed to automation? Attributions of responsibility for social problems help shape mass opinion. In this study I use two experiments, including one nationally representative probability survey-experiment, to examine the consequences of attributing job loss to trade versus automation. Findings suggest that as of 2018, public discourse attributes manufacturing job loss in America primarily to trade. When I experimentally manipulate attributions of responsibility for job loss, I find important consequences for levels of mass support for international trade, the extent of negative emotions arising from job loss, and beliefs that trade restrictions and tariffs can bring back manufacturing jobs. Finally, job loss that is attributed to trade—even when it is a single job loss—also serves as a threat to the national ingroup, which triggers a heightened sense of national superiority among white Americans.

Job loss can be devastating to those who experience it, regardless of its cause. Even for those who do not, rising unemployment can be a cause for collective concern about the state of the nation's economy and the prospects for an individual's own employment. In this study I ask what difference it makes that manufacturing job loss in the United States has been widely attributed to international trade rather than to automation. The answer to this question helps advance our understanding of the underlying bases of American trade preferences.

DIANA C. MUTZ is Samuel A. Stouffer Professor of Political Science and Communication at the University of Pennsylvania, Philadelphia, PA, USA. The author would like to thank Daniel Gurevitch and Annika Iyer for research assistance on this project. Data collection for this study was funded by National Science Foundation Grant #SES-1628057 to Time-sharing Experiments for the Social Sciences, Jeremy Freese (Stanford University) and James Druckman (Northwestern University), co-PIs. *Address correspondence to Diana C. Mutz, University of Pennsylvania, 133 S. 36th Street, Philadelphia, PA 19104-6215, USA; e-mail: mutz@upenn.edu.

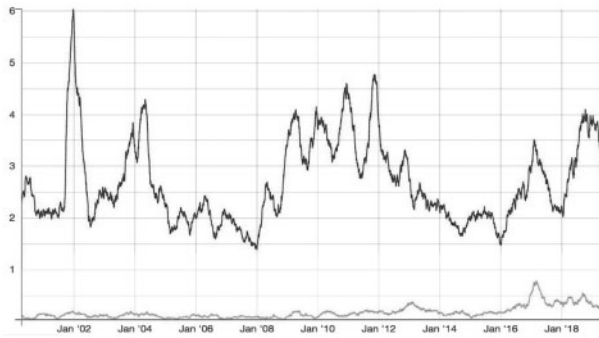


Figure 1. Newspaper articles linking job loss to trade and automation. Vertical axis represents counts of articles per day as a 7-day moving average. The solid black line represents articles linking job loss to trade and the gray line represents articles linking job loss to automation.

To date, studies of public opinion toward international trade have stressed either actual job loss or prospective concerns about personal or collective job loss as the root of public opposition (Scheve and Slaughter 2001; Mansfield and Mutz 2009). For example, the two most prominent theories of trade preferences suggest that individuals form trade preferences based on whether the person's industry of employment is import-competing or export-oriented, and whether they work at a low-skill or high-skill job (Margalit 2012).

Public debate likewise centers on the impact of trade on employment. According to patterns of news coverage, job loss in America is far more closely linked to trade than to automation. In figure 1, I use a massive data base of major US newspapers from 2000 to 2019 to plot the average number of articles per day mentioning job loss in conjunction with trade, and the average number mentioning job loss in conjunction with automation.¹ Although one might quibble as to whether all important synonyms for

1. To characterize public discourse on a large scale, I used all "Major US Newspapers" from the Nexis-Uni database, searching from 2000 to the present. Using a special license that allows large-scale bulk downloads of media content on weekends only, I downloaded all articles mentioning job loss or (un)employment and automation or trade. The exact search within Major US Newspapers was: ((job* w/10 (loss or declin*)) or unemployment) and (automat* or trad*). This search identifies any variant of the word "job" within ten words of loss or decline, or a reference to unemployment, but only those also mentioned in conjunction with trade or automation. I used a second program to search the downloaded collection of articles for those mentioning job loss along with automation-related terms, and a second search for those mentioning job loss along with references to trade. An article could thus include linkages to one or both topics at the same time. These counts were quantified by date. The specifications that created the two lines were:

Automation: (({job*} @50 ({loss} + {declin*})) + {unemployment}) & {automat*}

Trade: (({job*} @50 ({loss} + {declin*})) + {unemployment}) & {trad*}

automation or trade have been accounted for, the evidence overwhelmingly suggests that job loss is more frequently mentioned in close proximity to trade than to automation. Articles mentioning job loss in conjunction with automation hover near zero from 2000 through 2012, and then gradually increase over time, but they never reach even one-eighth of the prevalence of those mentioning job loss linked to trade. After 2016, such mentions increase, perhaps as a result of the 2016 election campaign. Americans clearly encounter trade discussed in relation to job loss far more than they encounter automation or technology discussed as a source of job loss. This study suggests that repeatedly linking job loss to trade as opposed to linking it to automation may have important consequences for mass opinion.

According to economists, whether trade flows increase, decrease, or have no net effect on the number of jobs in a country depends on a whole host of institutional characteristics. In other words, the relationship between international trade flows and aggregate unemployment rates is complex at best (e.g., [Belenkiy and Riker 2015](#)). Nonetheless, since the majority of Americans believe that trade causes job loss ([Guisinger 2017](#)), it seems logical that they oppose trade more when jobs are in short supply.

During the 2016 presidential election, job loss in the manufacturing sector became a major focus of public attention. Following precipitous and severe job losses beginning in the late 1990s, manufacturing employment in the United States actually *increased* from 2010 to 2016 ([Muro and Kulkarni 2016](#)), making 2016 an odd time for this issue to surface. Manufacturing job loss was also a surprising focus given the conventional wisdom among economists that most manufacturing job loss is due to automation rather than import competition. Contrary to political rhetoric on the left and right, a common view among economists is that most of the manufacturing jobs lost in the United States—roughly 85 percent—have been lost due to increased automation (see [Irwin 2016](#); [Hicks and Devaraj 2017](#); [Rose 2018](#)). Estimates vary, but they are comparable in suggesting that job loss due to automation is more threatening than offshoring ([Acemoglu et al. 2016](#); [Cocco 2016](#); [Miller 2016](#)).²

This study focuses on the consequences for public opinion of whether job loss is attributed to trade as opposed to automation, suggesting that misattributed blame for job loss may have important consequences for mass opinion (see [Wu 2018](#)). According to attribution theory, causal judgments are a central component of how people think about public issues; one's understanding of the underlying cause of a given problem determines his or her preferred policy solution ([Iyengar 1989, 1990](#)). However, in most such studies, the

2. Estimating what proportion of manufacturing jobs have been lost due to trade versus automation is not simple, thus this claim has not been without controversy, even among economists ([Houseman 2018](#); [Hicks and Devaraj 2017](#), [Chui, Manyika, and Miremadi 2016](#)).

distinction of interest is between internal versus external attributions of responsibility; in other words, is it the person's own fault they lost a job, or is it due to government policies or some external factor beyond the individual's control? Both automation and international trade policy are external phenomena that are beyond the individual's control. So why should it matter which is blamed for job loss?

Theory and Expectations

First, and most obviously, attributing responsibility for job loss to trade versus automation matters because seeing international trade as the culprit in this narrative should reduce public support for trade. While individual job loss has been shown to have little if any effect on individuals' trade preferences (e.g., [Rho and Tomz 2017](#)) or on policy preferences more generally ([Brody and Sniderman 1977](#); [Sniderman and Brody 1977](#)), sympathetic exemplars of larger societal problems have the capacity to alter public opinion ([Zillmann and Brosius 2000](#)).

Further, attribution theory suggests that pinning responsibility for job loss on foreigners as opposed to robots should result in significantly different reactions for two reasons. First, perceived intent plays an important role in attribution theory ([Heider 1958](#); [Kelley and Michela 1980](#)); in fact, intent sometimes matters more than outcomes to people's evaluations. People from other countries who "take American jobs" are more likely to be viewed as humans with intent than are robots who take jobs as a result of automation. When a person argues that "foreigners are stealing our jobs," this is an assertion of a purposeful, malicious act on the part of foreigners. Robots do not have intent and thus are less likely to be viewed in the same way. Thus, jobs lost to trade should elicit more anger than jobs lost to robots.

Third, these two attributions are likely to have different consequences because foreigners are considered outgroup members while robots are not. As [Rose \(2010\)](#) notes, "We don't think twice about trade between different states. Indeed, federal law prohibits anything that creates barriers to interstate trade. Should we think twice about trade between countries?" Indeed, Americans do not have the same aversion to interstate trade as they do to international trade ([Mansfield and Mutz 2013](#)). Whites in the United States are significantly more supportive of trade with predominantly white, English-speaking countries such as Canada and England than they are racially different countries such as China or Africa ([Sabet 2013](#)).

Beyond economic theories, the theoretical framework that has been most useful in understanding trade preferences involves the tendency for people to divide the world into ingroups and outgroups (e.g., [Mutz and Kim 2017](#)). Concepts closely related to in-group favoritism have been found to be related to trade preferences. For example, surveys have found trade opposition

related to personality characteristics such as authoritarianism (Johnston 2013), ethnocentrism (Mansfield and Mutz 2009), and a sense of national superiority (Rankin 2001; Mayda and Rodrik 2005; Margalit 2012). Those with high levels of racial prejudice also oppose trade (Sabet 2013). Although these are distinct concepts, they all represent tendencies toward in-group favoritism, that is, the tendency to think in rigid, “us-versus-them” terms.

Because foreigners are more likely to be viewed as outgroup members, attributing an American’s job loss to international trade should produce a sense of outgroup threat that job loss due to automation does not. What most theories of intergroup conflict share is a prediction that when a dominant ingroup’s power is threatened, ingroup members respond either by reasserting the superiority of their own group, and/or by derogating the outgroup (e.g., Tajfel and Turner 1979; Bobo and Hutchings 1996; Hutchings et al. 2010).

As outgroup members, foreigners are easy targets for blame and scapegoating (Tajfel and Turner 1986). For example, in a study of climate change policy attitudes, Americans who were told that climate change was being driven by excessive energy use by the Chinese were more likely to claim that climate change was within human control (Jang 2013). People naturally become defensive as a means of maintaining a positive ingroup identity. For this reason, attributions of responsibility for domestic job loss to a foreign entity should result in more negative views of outgroup members relative to ingroup members. Job loss due to trade allows citizens to personify the enemy-as-foreigner in a way that automation does not.

Attributing job loss to foreigners is likely to produce a defensive reaction among Americans high in ingroup favoritism. Threats to the ingroup generally strengthen ingroup identity (Sherif et al. 1961; LeVine and Campbell 1972; Giles and Evans 1985). As a result, attributions of responsibility for job loss to trade should promote more negative attitudes toward those in trading partner countries relative to Americans.

Further, if one believes that trade is responsible for job losses, then the solution to this problem is simple: change trade policies. Experimental studies suggest that the demand for protectionist policies increases in response to layoffs (DiTella and Rodrik 2019). But couldn’t government also “fix” job loss caused by automation? Because technology and automation are associated with progress, it seems unlikely that citizens would demand that government step in and prevent technological progress, even if it is responsible for the loss of jobs.

This combination of predictions rooted in attribution theory and theories of intergroup relations suggests five hypotheses about attributions of job loss to trade versus automation:

H1: Attributing job loss to trade will produce more negative emotions than when the same individual’s job loss is attributed to automation.

H2: Attributing job loss to trade will encourage more negative attitudes toward trade relative to when job loss is attributed to automation.

H3: Attributing job loss to trade will increase negativity toward the foreign outgroup and/or positivity toward the American ingroup relative to when job loss is attributed to automation.

H4: Attributing job loss to trade will encourage people to believe that manufacturing jobs can be brought back relative to when the same job loss is attributed to automation.

Additional hypotheses are tested using a control condition in which job loss is not attributed to either cause. Given the strong linkage between trade and job loss already present in public discourse, I expect that most people will assume that any job lost in manufacturing is likely to be due to trade, even when there is no information provided to this effect. The control condition indicates the current default interpretation, and thus allows inferences about how attitudes might change if different attributions of responsibility had been emphasized in public discourse.

Based on the theories described, those who identify strongly with the ingroup, that is, as Americans, should be most reactive to external threat. This study did not ask directly about the strength of people's identification as Americans before administering the experimental treatment because of the potential for artificially priming American identity. However, low levels of education are an excellent proxy for a strong American ingroup identity. Americans who have not attended college identify strongly as American; they also hold the most exclusionary conceptions of what it means to be American (Theiss-Morse 2009; Bonikowski and DiMaggio 2016; Jardina 2019). Thus, the threat from foreigners should serve to reinforce their sense of national superiority, whereas attributing job loss to technology should not.

H5: Attributing job loss to trade will produce stronger reactions among less educated Americans in bolstering their sense of American superiority.

Research Design

To examine these predictions, this study utilized a representative national probability sample made possible by Time-sharing Experiments for the Social Sciences (TESS). Data were gathered October 5–22, 2017, by the National Opinion Research Center at the University of Chicago/Amerispeak (see the [Supplementary Material, Appendix D](#), for details). The study design capitalized on the representativeness of a national probability survey sample, combined with the strong causal inferences that can be drawn from experimental data.

Since ingroup-outgroup dynamics affect dominant and subordinate groups differently (Tajfel 1982), the sample was limited to non-Hispanic white

Americans. Moreover, because attributions of responsibility for job loss due to trade often involve countries that are racially different from predominantly white America, a sample of Americans from the dominant white racial group holds constant the extent to which they will interpret job loss caused by trade in this experimental treatment involving China to be losses caused by racial outgroup members.

Over 1,000 white respondents were randomly assigned to one of three experimental conditions, each of which was identical in that it described a manufacturing worker who had lost his job. The experimental conditions varied only one thing: the attribution of responsibility for his job loss as 1) Unattributed job loss, 2) Job loss due to trade with China, or 3) Job loss due to automation.

Before launching the representative national survey experiment, I ran a pilot version of the larger experiment on a convenience sample obtained using Amazon's Mechanical Turk. The pilot study included just over 250 subjects divided into two conditions, job loss due to automation and job loss due to trade. The control condition was eliminated in order to reduce the need for subjects, with the underlying logic that if there were no differences found between those who were exposed to stories attributing a man's job loss to trade versus automation, then a control condition would be unlikely to reveal additional information. The Pilot Study questions varied only slightly from the items used in the final study (see the [Supplementary Material, Appendixes A and B](#)).

The experimental treatment used in both the pilot and the final study was drawn from a front-page article in the *New York Times*.³ Like many news stories about job loss, it featured a man who had recently lost his manufacturing job:

When Michael Morrison took a job at the steel mill in the center of Granite City, Ill., in 1999, he assumed his future was ironclad. He was 38, a father with three young children.

"I felt like I had finally gotten into a place that was so reliable I could retire there," he said.

Although it had changed hands, the mill had been there since the end of the 19th century. For those willing to sweat, the mill was a reliable means of supporting a family.

Mr. Morrison began by shoveling slag out of the furnaces, working his way up to crane driver. From inside a cockpit tucked in the rafters of the building, he manned the controls, guiding a 350-ton ladle that spilled molten iron.

3. <https://www.nytimes.com/2016/09/29/business/economy/more-wealth-more-jobs-but-not-for-everyone-what-fuels-the-backlash-on-trade.html?searchResultPosition=5>.

It was a difficult job requiring perpetual focus and he was paid accordingly.

Only one part of the article varied across the three experimental conditions. In the control condition, readers learned simply that his job was eliminated, but they were not told why:

Now his job has been eliminated.

In the trade-related job loss condition, they were told that his job was eliminated due to Chinese trade competition:

Now his job has been eliminated due to trade with China. Chinese workers now man the same machine that Mr. Morrison once operated. As the company website describes, “Of the 74 machines that were operating in the factory, 63 are now operating in China.”

In the automation-related job loss condition, Mr. Morrison was replaced by a robot:

Now his job has been eliminated due to automation. Robots now man the same machine that Mr. Morrison once operated. As the company website describes, “Of the 74 machines that were operating in the factory, 63 now run on their own with no human intervention.”

All versions of the article ended identically as follows:

Mr. Morrison has not been able to find other work, and he has no idea how he will pay for his children’s college educations. “When they don’t need me anymore,” he said, “I’m nothing.”

Highlighting a single individual might seem unlikely to generate strong reactions given that it is just one person who lost one job. But numerous studies have demonstrated that the power of single identifiable victims easily outstrips unidentified statistical victims (Lyon and Slovic 1976; Fetherstonhaugh et al. 1997; Small and Loewenstein 2003; Kogut and Ritov 2005; Small, Loewenstein, and Slovic 2007; Small and Lerner 2008; Small 2010). More emotionally engaging portrayals create greater sympathy for victims and mediate effects on policy attitudes (Gross 2008). Since the man loses his job across all conditions, the negative tone of the story and the extent of job loss remains constant.

Immediately after reading the article, respondents were asked for their emotions in response to the story. On a five-point agree-disagree scale, they rated the extent to which they agreed or disagreed that the article made them sad, angry, resentful, or irritated. These items were combined into an additive scale of Negative Emotions (Cronbach’s $\alpha = 0.77$).⁴

4. In the pilot study, only two negative emotions were measured (see the [Supplementary Material, Appendix A](#)).

Four questions were used to tap attitudes toward trade and globalization.⁵ They included whether people thought that government should try to encourage international trade or try to discourage international trade, whether the respondent believes that globalization is good or bad for the United States, whether they favor or oppose the federal government in Washington negotiating more free trade agreements, and whether the increasing amount of trade between the United States and other countries has helped or hurt the US economy. In both studies, the items produced a highly reliable index of support for Trade/Globalization (Pilot Cronbach's $\alpha = 0.83$; National Sample Cronbach's $\alpha = 0.81$).⁶

When measuring the expression of negative attitudes toward specific groups, researchers are often concerned that respondents will not want to state explicitly negative views of outgroups due to social desirability. In the Pilot Study, I asked about attitudes toward the United States and China in two ways. First, I asked them to evaluate Americans and Chinese separately on scales running from Trustworthy to Untrustworthy (see the [Supplementary Material, Appendix A](#)). I then computed a measure of how much more trustworthy Americans perceived themselves to be relative to the Chinese. This parallels the construction of other common measures of ethnocentrism and ingroup favoritism (Kinder and Kam 2010).

In both experiments, I also asked respondents to evaluate a series of statements that might (or might not) be considered acceptable in a social situation. Regardless of their views of Asians, people may be reluctant to endorse stereotypes themselves, so they were instead asked whether other people who made various statements were expressing socially unacceptable views. These statements derogated Chinese people or Asians in general relative to Americans using common stereotypes of Asians (see the [Supplementary Material, Appendix B](#)). The list was the same in both experiments, and it included derogatory comments about Chinese people as well as assertions of American superiority to China. These items were used to create an index of the Acceptability of Asserting American Superiority to the Chinese by taking the mean, with all variables coded so that high scores indicated more American superiority relative to the Chinese. Both studies produced highly reliable indexes (Pilot Cronbach's $\alpha = 0.79$; National Sample Cronbach's $\alpha = 0.85$). My hypothesis based on how threat is known to affect ingroup/outgroup attitudes is that these negative stereotypes should be deemed more socially acceptable among respondents recently reminded by the article of manufacturing job loss due to trade with China.

5. In the pilot study, only three of these four questions were asked (see the [Supplementary Material, Appendix A](#)).

6. See the [Supplementary Material, Appendix B](#), for question wording of items in the National Survey Experiment that are not included in the text.

Belief that Manufacturing Jobs Could be Returned to the US was measured by asking respondents to agree or disagree with a statement asserting, “The manufacturing jobs that have left the US will come back if we put taxes or tariffs on foreign goods coming into the US.”⁷ Respondents also estimated the percentage of manufacturing jobs lost that were attributable to trade (see the [Supplementary Material, Appendix A and B](#)).

At the very end of the study, after measuring all other variables, a manipulation check assessed whether respondents had read the article carefully enough to notice the cause of job loss that was suggested. Given sometimes fleeting attention in online surveys, and the fact that this manipulation was subtle, mentioning the cause in only one paragraph, manipulation checks provide a means of ensuring that respondents read the story in full and were aware of the cause of the man’s job loss. Respondents were asked why the man in the story lost his job, and given four options: a) his job was sent to China, b) he was replaced by new technology/robots, c) his company merged with another company so he was no longer needed, and d) the article said nothing about why he lost his job. This question was asked last in the study so as not to call undue attention to the cause of the man’s job loss before the dependent variables were assessed. The manipulation checks in both studies indicated that respondents understood the experimental treatments as intended (see the [Supplementary Material, Appendix C](#), for details). All subjects were retained in the analyses in order to preserve random assignment.⁸

Results

PILOT STUDY

To analyze the pilot study results, I compared the means for each of the measures corresponding to Hypotheses 1 through 4 after adjusting for Isolationism. The results in [table 1](#) are largely consistent with my predictions, though not conclusive on every single measure. Hypothesis 1, predicting more Negative Emotions among those reading about job loss due to trade, was only marginally supported. Hypothesis 2, predicting that support for trade would be greater when job loss was framed as a result of automation as opposed to trade competition, was supported. On average, those

7. Pilot Study respondents were asked a slightly different question tapping the same construct: “If the right policies are put in place, it will be possible to increase the number of manufacturing jobs in the US back to what they were before.”

8. In the analyses of the Pilot Study, an index tapping Isolationist attitudes was measured as a pre-treatment covariate and included to increase the efficiency of the model given its small sample size (see the [Supplementary Material, Appendix A](#)). Isolationism has a well-established influence on trade attitudes ([Mansfield and Mutz 2009](#)).

Table 1. Results for pilot experiment using Mechanical Turk sample

	Job loss due to trade		Job loss due to automation		<i>p</i> -value for mean difference
	Mean	(se)	Mean	(se)	
Negative emotions	0.63	(0.02)	0.58	(0.02)	0.062
Support for trade	0.54	(0.02)	0.59	(0.02)	0.044
Believe jobs could return	0.69	(0.03)	0.61	(0.03)	0.015
Acceptability of asserting American superiority	0.32	(0.02)	0.24	(0.02)	0.001
Trustworthiness of US versus China	0.51	(0.02)	0.46	(0.02)	0.012
Trustworthy US	0.67	(0.02)	0.55	(0.02)	0.000
Trustworthy China	0.47	(0.02)	0.45	(0.02)	0.280
Trade versus automation at fault	0.45	(0.02)	0.36	(0.02)	0.000
Sample <i>n</i>	125		126		

NOTE.—Entries are means and standard errors by experimental condition. All variables are on a 0 to 1 scale. *p*-values are for directional predictions (one-tailed tests).

exposed to the story mentioning automation as responsible were 5 percent more favorable toward trade.

To what extent did feelings about the superiority of Americans change in reaction to a story about job loss due to trade relative to a story about job loss due to automation? Hypothesis 3 was that those threatened by international trade would respond by asserting greater American superiority than would those in the automation condition, where ingroup and outgroup attitudes would not be triggered. The index tapping whether derogatory statements about Chinese people relative to Americans were socially acceptable demonstrated a clear difference by experimental condition. Those randomly exposed to job loss due to trade were far more likely to say that stereotypes asserting American superiority to the Chinese were socially acceptable. In short, negative stereotypes were deemed significantly more appropriate among those who read about job loss due to trade than among those who read about a job loss due to automation.

Respondents who read about job loss due to competition with China also reported a greater gap between the perceived trustworthiness of Americans relative to Chinese. As shown in table 1, although Americans routinely perceive themselves as more trustworthy, job loss due to trade resulted in a 5 percent boost in perceptions of superior trustworthiness. When that measure was broken down so one could look at the perceived trustworthiness of Americans and Chinese separately, it becomes apparent that American

respondents did not downgrade their views of the Chinese so much as they bolstered their ingroup positivity, asserting even greater trustworthiness among Americans. Consistent with a threat to ingroup identity, respondents asserted American superiority to an even greater degree.

What is most interesting about these results is that they suggest a reciprocal causal relationship between Americans' sense of national superiority and their attitudes toward trade. Past studies have found empirical relationships between a sense of national superiority and trade opposition, with perceived national superiority assumed to be the cause of trade opposition (e.g., O'Rourke and Sinnott 2002; Mayda and Rodrik 2005). However, when viewed through the lens of theories of intergroup relations, it also makes sense that those who feel threatened by trade should develop a stronger sense of national identity as a result. To believe that one's own national ingroup has more of a rightful claim to certain jobs than other countries, a person must believe that his or her national ingroup is superior to others'. Thus, the relationship between national superiority and threats from globalization may be mutually reinforcing.

Consistent with Hypothesis 4, respondents randomly assigned to the trade treatment were significantly more likely to believe that manufacturing jobs could come back than were those who read about job loss due to automation, with an 8 percent difference between the two treatments. As also illustrated in table 1, the relative importance of trade versus automation in creating job loss was also significantly different by condition, despite the fact that the treatment involved only a single case of job loss.

National Experiment. The nationally representative survey experiment provided an opportunity to replicate these findings with a larger and more diverse sample. By using varied operationalizations of the outcome measures, I tested the generalizability of these findings across operationalizations as well as on a representative national probability sample that makes effect sizes and heterogeneity of effects of interest.

Figure 2 shows that when comparing the conditions in which the man's job loss was attributed to trade as opposed to automation, the four main hypotheses were consistently supported. The extent of negative emotions was 11 percent lower when job loss was attributed to automation as opposed to trade (H1, $p < 0.001$). Support for trade was 6 percent higher when job loss was attributed to automation as opposed to trade (H2, $p < 0.001$). Assertions of American superiority to the Chinese were 6 percent higher when job loss was attributed to trade relative to automation (H3, $p < 0.01$). Likewise, those assigned to the trade condition were 6 percent more likely to believe that manufacturing jobs could be brought back with the right government policies (H4, $p < 0.01$). These findings are consistent with results from the Pilot Study in supporting the four main hypotheses, and they provide supportive evidence across a nationally diverse sample.

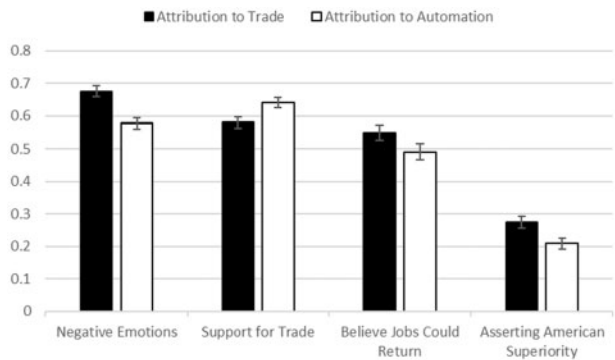


Figure 2. Effects of attribution of job loss to trade versus automation, based on data from the national experiment. In each pair, the means differences are statistically significant using two-tailed tests.

The national study included a control condition in which respondents were not given an explanation for the job loss of the man in the article. This allowed me to assess the effects of these alternative attributions independent of one another. The analyses in [table 2](#) compare each of the two treatment conditions, job loss due to automation and job loss due to trade, to the control condition using two dummy variables in a regression equation with the control condition as the reference category.

The coefficients in [table 2](#) indicate whether each treatment was different from the control condition rather than from the opposing attribution. The positive and negative signs of the coefficients consistently suggest that the control condition is in the middle relative to the conditions involving attribution to trade and attribution to automation, just as one would theoretically expect. As shown in Model 1, relative to a control condition involving the same man losing the same job without saying why, job loss due to trade produced significantly stronger negative emotions among experimental subjects. Respondents who read about job loss due to trade registered 9 percent stronger Negative Emotions compared to those in the control condition. Job loss due to automation produced no such effects. Given that even the control condition involved job loss, it is clear that these negative emotions are not simply a response to a man losing a job. In other words, it is not the sadness of job loss per se to which people are reacting. It is job loss due to trade that makes people angry in a way that job loss due to automation does not.

In Model 2 of [table 2](#), attributing one man’s job loss to automation produced significantly more support for international trade relative to the control condition. Attributing job loss to trade had no mirror impact, but this is likely because people already associated job loss strongly with trade. Explicitly attributing this man’s job loss to automation raised people’s support for trade

Table 2. Effects of job loss due to trade versus automation with reference to control condition, national experiment

	Model 1: Negative emotions			Model 2: Support for trade		
	B	Std Err	p-value	B	Std Err	p-value
Due to automation	-0.011	0.017	0.531	0.047	0.016	0.004
Due to trade	0.087	0.017	0.000	-0.011	0.016	0.489
Constant	0.588	0.012	0.000	0.592	0.012	0.000
Sample <i>n</i>	1,007			1,008		

	Model 3: Believe jobs could return			Model 4: Asserting American superiority		
	B	Std Err	p-value	B	Std Err	p-value
Due to automation	-0.042	0.024	0.080	-0.027	0.017	0.107
Due to trade	0.019	0.024	0.439	0.039	0.017	0.022
Constant	0.533	0.017	0.000	0.235	0.012	0.000
Sample <i>n</i>	1,009			1,009		

NOTE.—Entries are OLS regression coefficients representing random assignment to experimental attribution treatments. Baseline is control group.

by 5 percent. Whether this seems like a lot or a little impact is a matter of perspective. But it is worth noting that other experimental treatments—for example, suggesting that trade leads to lower prices for consumers—have not produced a significant positive impact on trade attitudes (Hiscox 2006).

Model 3 in table 2 assesses change in the extent to which people believe manufacturing jobs will come back if US policies are changed. Here neither coefficient achieves standard levels of statistical significance, although the marginally significant effect of the automation cue suggests roughly the same size of impact as it had on trade preferences.

In Model 4 of table 2, I assess the effects of these cues on Asserting American Superiority over Chinese people. Here my hypothesis was that attributing responsibility for job loss to trade (i.e., a foreign threat) should increase the extent to which Americans feel a need to assert their own superiority over other nationalities, in this case, especially the Chinese. Hearing about job loss due to trade should increase the sense of ingroup threat. The fourth model provides support for this prediction. Receiving the version of the story that attributes the man's job loss to trade increased the extent to which people thought it was acceptable to claim American superiority to the Chinese. The coefficient suggests a 4 percent increase in assertions of American superiority and relative derogation of Chinese people. Likewise, relative to the control condition, those who received the story attributing responsibility for the man's job loss to automation felt it was less appropriate to claim American superiority.

As a follow-up to my main hypothesis tests, for Hypothesis 5 I examined whether these effects were heterogeneous with respect to education, the variable most closely tied to national identity as well as to trade support (Hainmueller and Hiscox 2006). As shown in the [Supplementary Material, table C.1](#), education conditioned the impact of attributions as predicted in all four analyses. But most importantly, and consistent with my hypotheses about why trade and nationalistic views are closely related, the effects of attribution cues on the perceived acceptability of claiming American superiority were significantly larger for those without a college education. As shown in [figure 3](#), the largest effects were among those with the strongest attachment to American national identity. This finding lends further support to the idea that nationalistic attitudes can be a response to perceived threat to the national ingroup such as that perceived to be posed by trade. In other words, nationalistic beliefs are not only a potential cause of protectionist views. For those who feel America is threatened by international competition, trade competition produces stronger assertions of American superiority. Simply changing the attribution for a single man's job loss from trade to automation produced a 9 percent difference in American Superiority. Among the less educated, an attribution of job loss to automation reduced the extent to which

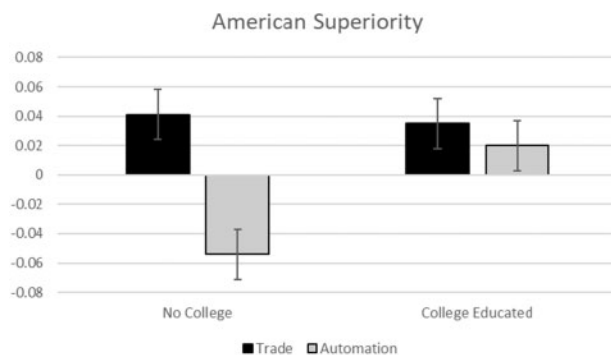


Figure 3. Interactions between college education and attribution cues in influencing perceived acceptability of American superiority, relative to control baseline. Means are adjusted so that the control condition mean is represented by zero on the y-axis. Bars represent deviations from the control condition due to trade-related and automation-related attributions. Details are provided in the [Supplementary Material, section C, table C.3](#).

China-bashing seemed appropriate and decreased the acceptability of asserting American superiority.

Limitations

The causal inferences drawn in this study are strong in most respects, although they include important limitations. The results rest on evidence from two independent experiments that produced consistent findings suggesting that attributions of responsibility for job loss matter to trade preferences. Given that the second experiment involved a representative national sample of whites, its results are highly generalizable to the non-minority American population. Because the studies utilize random assignment to conditions, they facilitate strong causal inferences. Moreover, by relying on multiple-item measures of outcomes, the operationalizations have greater reliability than single-item measures. Given that operational measures varied, one can have greater confidence that the findings do not depend on any specific operationalization.

The main limitations of this study stem from the fact that both are one-shot experiments involving exposure to a single stimulus. I do not assess the persistence of the observed effects over time. At the same time, the fact that a single short treatment had such an impact on four outcomes—emotions, trade preferences, beliefs that manufacturing jobs could come back, and assertions of American superiority—is somewhat surprising. The experimental treatment says nothing about job loss as a larger-scale phenomenon or

about whether the attribution for this one man's job loss is generalizable to others who have lost jobs. Nonetheless, the story altered people's assessments of the relative importance of trade and automation for job loss, thus altering multiple dimensions of their judgments about international trade.

The all-white sample also limits the generalizability of these results. One would not necessarily expect these findings to generalize to minorities in the United States, given that they are both outgroup members within the United States, and ingroup members as Americans. Further, because they are ingroup members with lower levels of national attachment, they are less likely to react defensively by elevating their sense of American superiority in response to hearing about a person who loses a job to someone overseas, and less likely to be angered by jobs going overseas (see [Mutz, Mansfield, and Kim 2021](#)).

Given that this was an experiment, external validity is also a concern. How likely is it that people's assessments of the causes of job loss are induced to vary in the real world as they did in this representative national survey experiment? As described above, this specific story came almost verbatim from the *New York Times*. But how common are such stories? The dominant frame in media coverage of unemployment is an individual person's job loss ([Reese, Daly and Hardy 1987](#); [Guisinger 2017](#)). Moreover, stories about job loss typically draw on specific examples of a sympathetic individual that has lost his or her job.⁹ The stories are often gut-wrenching. Studies of attribution suggest that emotionally charged attributions of responsibility such as this man's story are particularly potent ([Hameleers, Bos, and de Vreese 2017](#)).

A final limitation of the study stems from my exclusive focus on American discourse on trade and its consequences for American respondents. Manufacturing job loss has declined in other industrialized countries as well as in the United States. This research cannot speak directly to whether similar misattributions of responsibility have occurred elsewhere. However, the psychological underpinnings of people's reactions to job loss when they believe jobs are going to foreigners should be very similar. Ingroup-outgroup dynamics are common across international borders.

Conclusion

These findings further our understanding of the psychological underpinnings of mass opinion toward trade. Although job loss is clearly an event of great consequence, the mass public is reacting to more than the economic consequences of job loss when forming attitudes toward international trade. The

9. Positive coverage, in contrast, tends to portray job gains in terms of abstract statistical trends such as increasing economic growth or exports, rather than in explicitly human terms.

psychology underlying American trade preferences also draws on people's tendencies to divide the world into ingroups and outgroups, a competitive contest involving "us versus them." Job loss due to trade threatens people's ingroups in a way that job loss due to automation does not.

If, as [Irwin \(2018\)](#) has argued, when it comes to job loss, "The main culprit is technology," then extensive public discourse emphasizing trade as a source of job loss rather than automation has had important consequences for American public opinion. For one, it has made people angrier. Job loss due to trade promotes significantly more negative emotions than job loss due to automation. Angry reactions to trade are not simply about job loss. Although job loss is obviously an enormously negative event, what makes people far more upset is the idea that other people are benefiting from their own bad fortune. When those other people are foreigners, it is especially upsetting. Casting foreign countries as the villains in these narratives has consequences that go beyond potentially misperceiving the main source of the problem. It also encourages ethnocentrism and xenophobia. At the policy level, it tilts the playing field toward mercantilist views and more isolationist foreign policy.

These attributions of responsibility have sizable effects. While control conditions are useful for purposes of understanding the independent impact of attributions to trade and automation, what is most generalizable to the real world is the difference between stories that attribute trade to one cause versus the other. Most stories about job loss attribute a cause to either trade or automation. When journalists make a choice about whom to feature when discussing manufacturing job loss, this choice is not without consequence. This begs the question as to why the linkage between job loss and trade is so much more common than for automation. I can only speculate, but automation is not as politically controversial an issue. This makes it difficult for journalists to cover it as a battle between conflicting political forces. In contrast, a conflict between "us" and "them" provides a compelling storyline.

The size of these effects is striking given the subtlety of the treatment. In a representative sample of Americans, among those most anti-trade, that is, those without a college education, trade support increased by 6 percent after reading about a single job loss due to technology. This finding begs the question of why these seemingly subtle differences in a single article might successfully increase levels of trade support when far more involved argumentation often does not. When economists have tried to increase Americans' support for international trade, they have typically turned to economic arguments, suggesting, for example, that the prices consumers pay will decline as a result of international trade. Such efforts have not been successful (see [Hiscox 2006](#)).

Attributions of responsibility, on the other hand, are a common component of everyday reasoning. Further, these attributions of responsibility seem less

likely to prompt counter-arguments than direct persuasive argumentation does. It is relatively easy for a person confronted with the argument that trade saves them money to counter that this is not what is most important to them, or that they do not believe this to be the case. On the other hand, when a story says that a man lost his job due to trade (or automation), it seems unlikely that readers should doubt this claim. Believing this claim does not, on its face, ask them to change their views about trade. Nonetheless, prompting them to think about this sympathetically portrayed job loss as caused by automation versus trade has significant downstream effects.

Finally, these findings also shed light on the current alignment of nationalism, xenophobia, and trade opposition. Nationalism is often acknowledged as a cause of trade opposition. People who believe their country is superior to other countries are less likely to want to do business with those other countries. But it is worth considering, in addition, an alternative mechanism by which nationalism and opposition to international trade become close bedfellows. When people attribute job loss to foreigners taking Americans' jobs, the expected psychological response to this threat is for people to become more positive toward their national ingroup while actively denigrating the outgroup. Thus nationalism, xenophobia, and trade opposition go hand in hand in a way that nationalism and job loss due to automation do not.

Data Availability Statement

REPLICATION DATA AND DOCUMENTATION are available at Mutz, Diana, 2020, Replication Data for “(Mis)Attributing the Causes of American Job Loss: The Consequences of Getting It Wrong,” <https://doi.org/10.7910/DVN/RZPZYV>, Harvard Dataverse.

Supplementary Material

SUPPLEMENTARY MATERIAL may be found in the online version of this article: <https://doi.org/10.1093/poq/nfab003>.

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